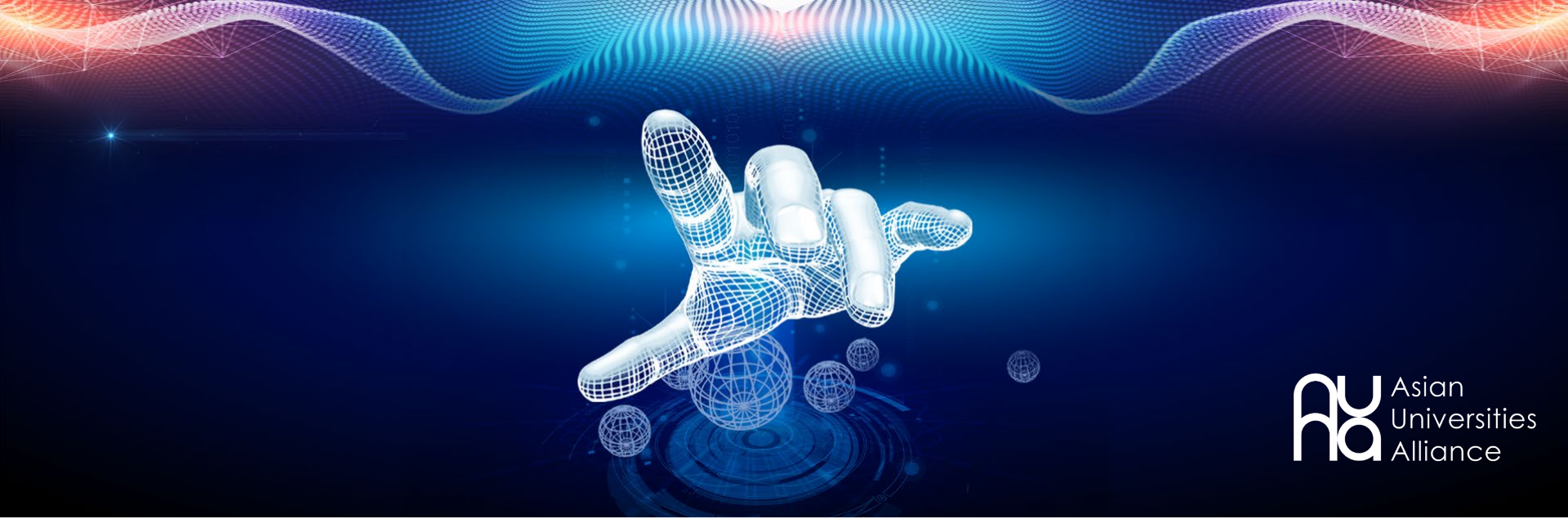

LEARNING SPARSE STRUCTURED ENSEMBLES WITH STOCASTIC GTADIENT MCMC SAMPLING AND NETWORK PRUNING

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This work is accepted by IEEE International Workshop on Machine Learning and Signal Processing, 2018.

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2 Related Work

3 Our solution: Mix of multiple ingredients

- Learning ensembles via SG-MCMC sampling
- Cost reduction via structured model compression
- Experimental results

4 Conclusion & Future Work

Ensemble of Neural Networks

- Ensemble models are a **group of models** that work collectively to get the averaged prediction.

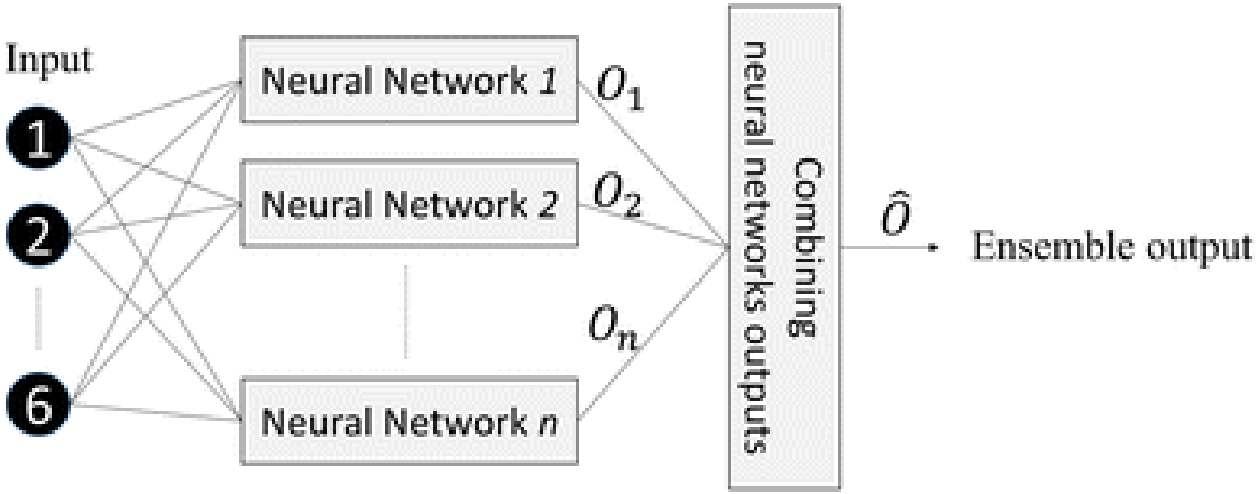


Figure from Effat Dehghanian et al. 2015

Ensemble of Neural Networks

- Ensemble gives a **great boost in accuracy** because it does not rely on a single model for prediction.

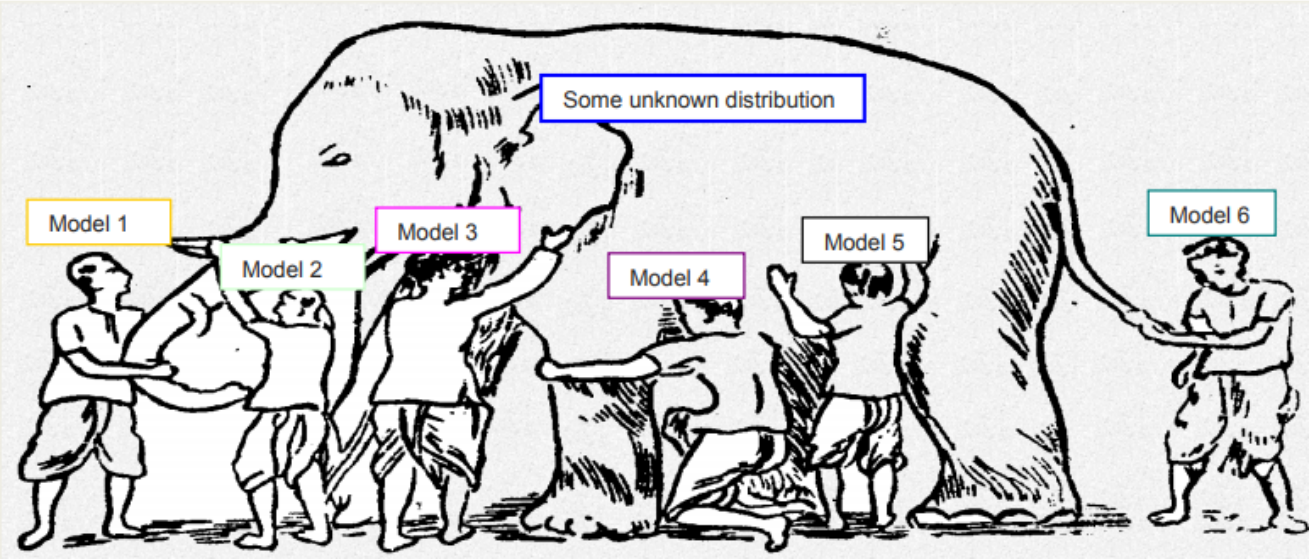
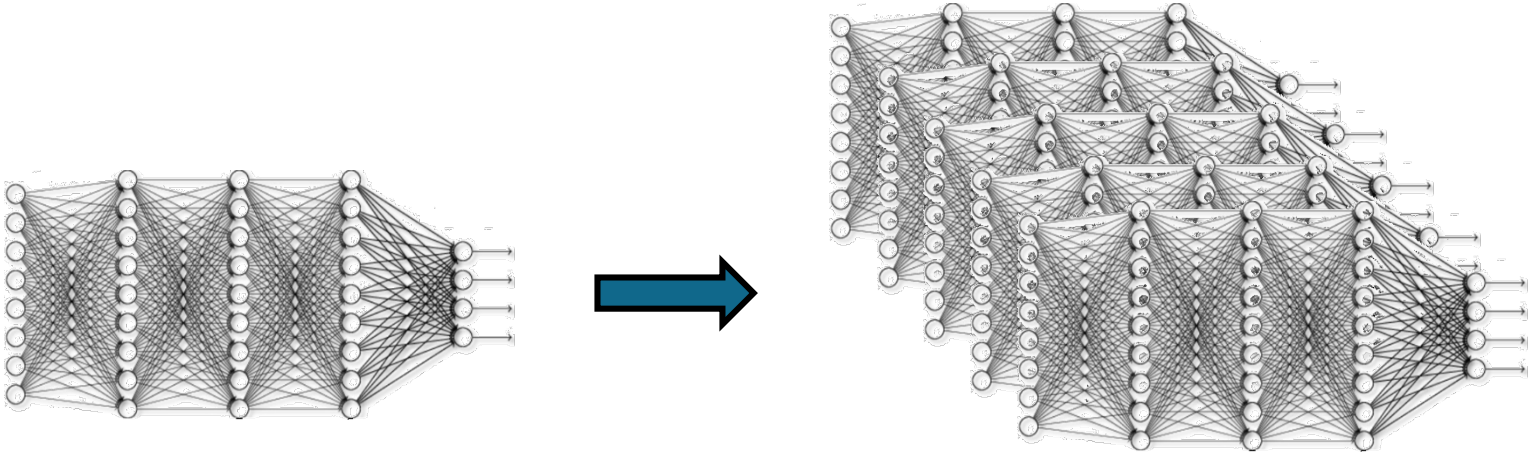


Figure from Alejandro Correa et al. 2013

Problem



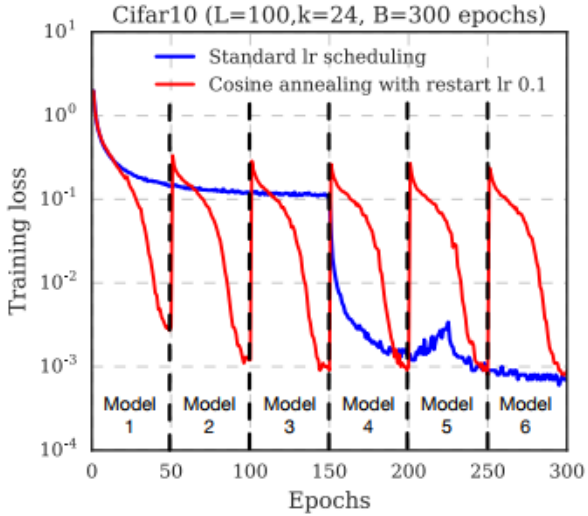
N networks

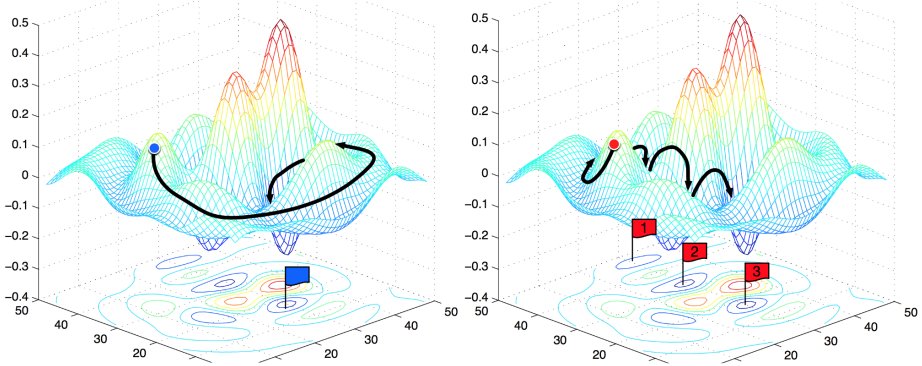
- **Training problem:** N times training time
- **Testing problem:** N times memory/testing time cost

Related Work

Snapshot ensembles: Train 1, get m for free (Gao Huang et al. 2017)

- Obtain multiple snapshot models within a single training process.
- Empirical cyclic learning rate settings.





Related Work

- The recent progress in Bayesian posterior sampling:

  Stochastic Gradient Markov Chain Monte Carlo sampling algorithms

  (Max Welling et al. 2011, Tianqi Chen et al. 2014, Zhe Gan et al. 2016)

- SG-MCMC works by adding a scaled gradient noise to Stochastic optimization method which is proved to have the following benefits :

  (i) Theoretically interpretable

  (ii) Efficient exploration of the model parameter space

  (iii) Scalable and simple

Related Work

- Testing problem: N times memory/testing time cost

Model compression via
Network pruning and retraining
(Song Han et al. 2015, 2017).

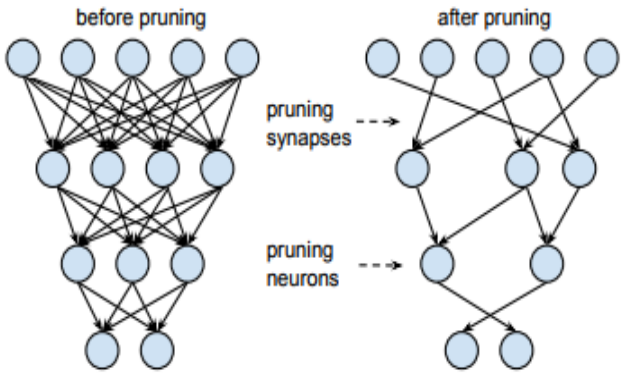


Figure from Song Han et al. 2015

Sparse structure learning via
Group Lasso penalty (Ming Yuan et al. 2006) on
deep models (Wei Wen et al. 2016, 2017).

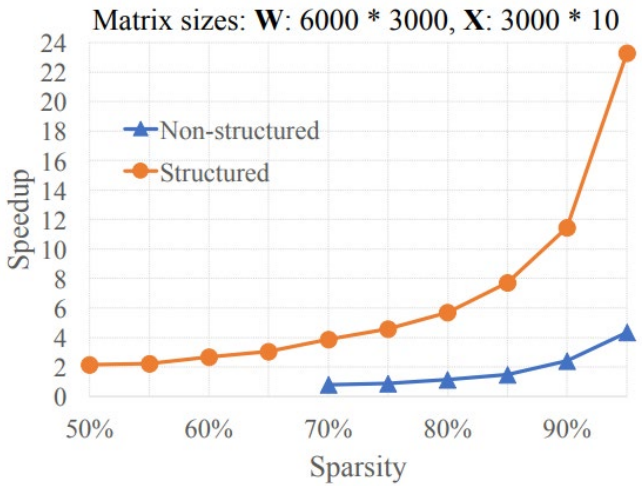


Figure from Wei Wen et al. 2017

Our Work

SG-MCMC based Bayesian learning + **Group Sparse Prior** + **Network Pruning & Retraining**

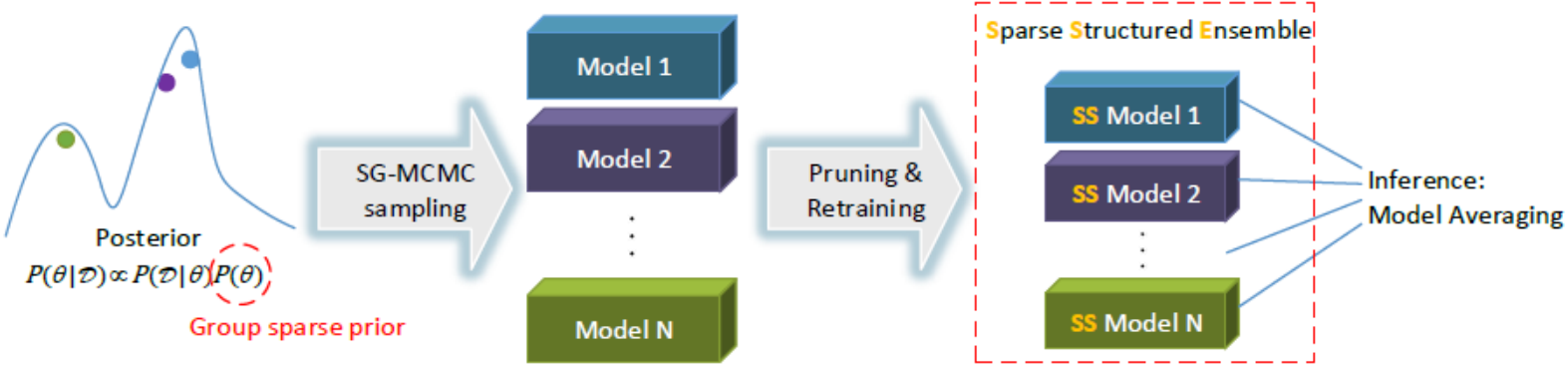


Figure 1: Overview of our two-stage method for learning SSEs.

Training: SG-MCMC Sampling

- $p(\tilde{y}|\tilde{x}, D) \approx \frac{1}{M}\sum_{m=1}^{M} p(\tilde{y}|\tilde{x}, \theta_m) \quad ,\theta_m \sim p(\theta|D)$
- Goal of training: sample $\theta \sim p(\theta|D)$, obtain $\{\theta_m\}_{m=1}^{M}$ via SG-MCMC

Stochastic Gradient Langevin Dynamic (Welling and Teh, 2011, the simplest SG-MCMC):

$$\tilde{g}_t = \nabla \log p(\theta_t) + \frac{N}{n}\sum_{i=1}^{n} \nabla \log p\left(y_t^{(i)} \middle| x_t^{(i)}, \theta_t\right),$$

$$\Delta\theta_t = \epsilon_t \tilde{g}_t + \eta_t, \qquad \eta_t \sim \mathcal{N}(0, 2\epsilon_t)$$

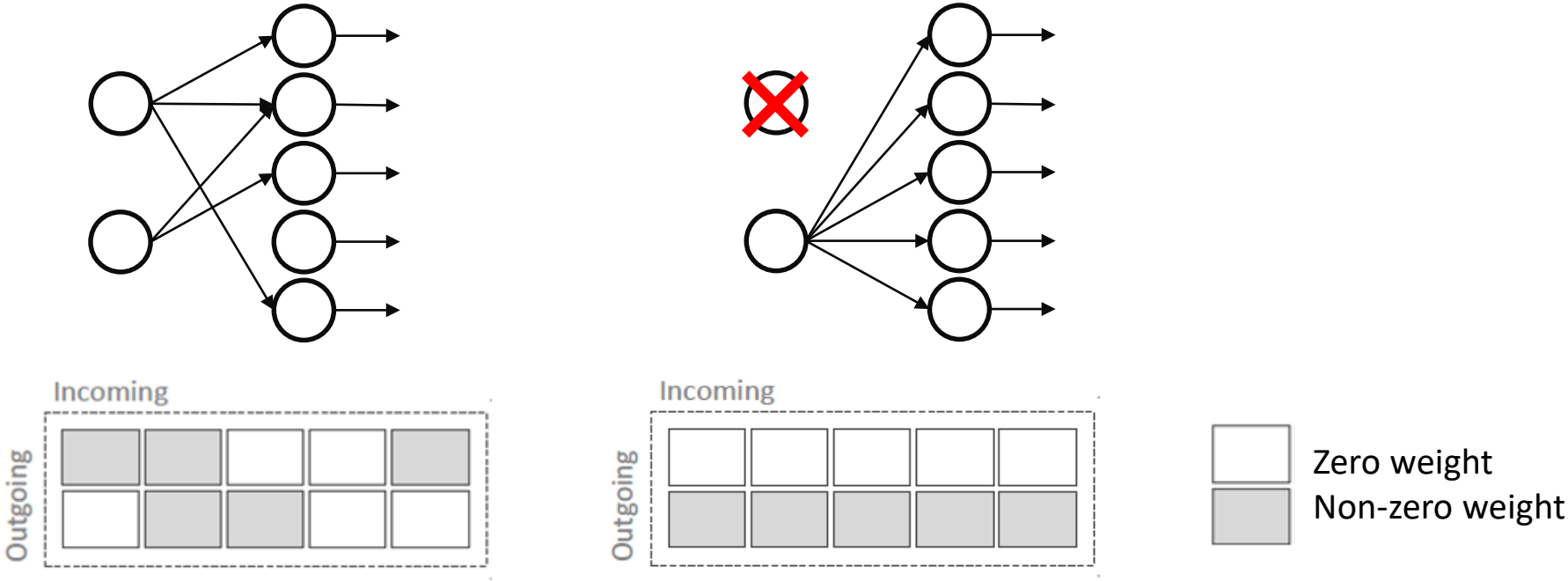
Compared to Stochastic Gradient Descent:

$$\tilde{g}_t = \frac{N}{n}\sum_{i=1}^{n} \nabla \log p\left(y_t^{(i)} \middle| x_t^{(i)}, \theta_t\right),$$

$$\Delta\theta_t = \epsilon_t \tilde{g}_t$$

Sparse Structured FNN

- Pruning of Fully-connected Neural Networks



Sparse Structured LSTM

- Pruning of LSTMs

$$
\begin{aligned}
\boldsymbol{f}_t &= \sigma([\boldsymbol{x}_t, \boldsymbol{h}_{t-1}]W_f + \boldsymbol{b}_f) \\
\boldsymbol{u}_t &= \tanh([\boldsymbol{x}_t, \boldsymbol{h}_{t-1}]W_u + \boldsymbol{b}_c) \\
\boldsymbol{c}_t &= \boldsymbol{f}_t \odot \boldsymbol{c}_{t-1} + \boldsymbol{i}_t \odot \boldsymbol{u}_t \\
\boldsymbol{i}_t &= \sigma([\boldsymbol{x}_t, \boldsymbol{h}_{t-1}]W_i + \boldsymbol{b}_i) \\
\boldsymbol{o}_t &= \sigma([\boldsymbol{x}_t, \boldsymbol{h}_{t-1}]W_o + \boldsymbol{b}_o) \\
\boldsymbol{h}_t &= \boldsymbol{o}_t \odot \tanh(\boldsymbol{c}_t)
\end{aligned}
$$

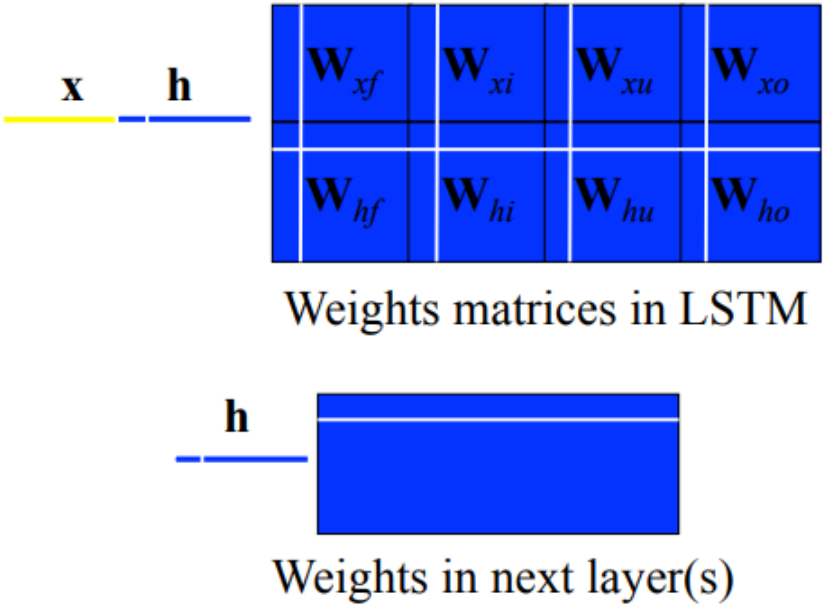
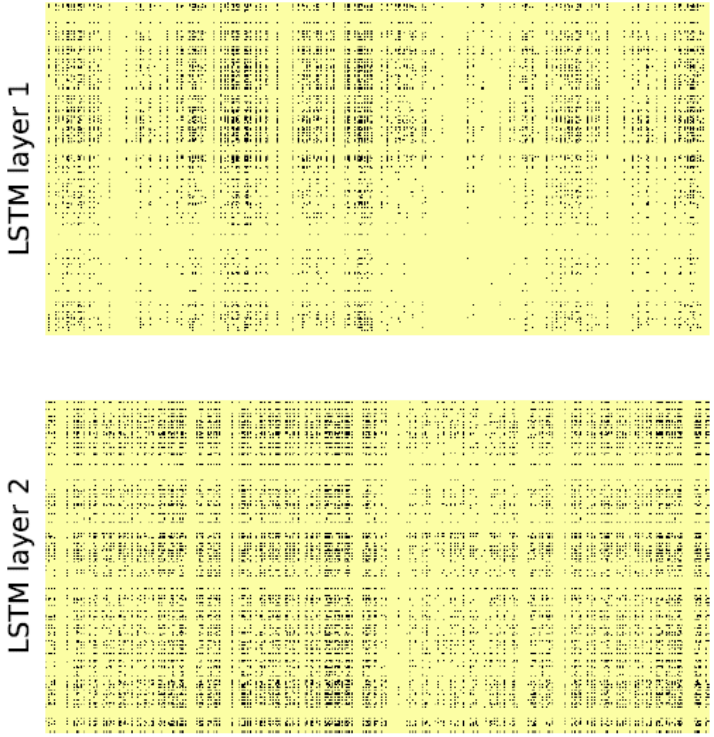


Figure from Wei Wen et al. 2017

Sparse Structured LSTM



	Structure	Params	FLOPs
Original model	1500-1500-1500	51M	100%
Pruned model	533-425-533	9M(18%)	18%

FLOPs for a matrix *W* is calculated as the size of the smallest sub-matrix formed by such rows and columns that contain all non-zero elements in *W.*

Language Modeling Experiment

- Perplexity on PTB dataset

Method	Model	Params	FLOPs	Dev.	Test
SGD [10]	1 large	1*	1*	82.2	78.4
SGD [10]	38 large	38×	38×	71.9	68.7
VD [24]	10 large	10×	-	-	68.7
VD+SEAL [11]	individual	51M	-	71.1	68.5
SGLD+GSP+PR	20 large	2.0×	4.5×	68.6	66.4
SGLD+GSP+PR	4 large	0.4×	0.5×	72.2	69.7
SGLD+GSP+PR+SE	4 large	**0.3×**	**0.7 ×**	**64.4**	**62.1**

* The baseline LSTM model has 66M parameters and 102M FLOPs.

Ref： [10] Wojciech Zaremba et al. 2014; [11] Hakan Inan et al. 2017; [24] Yarin Gal et al. 2016

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Conclusion & Future Work

Conclusion:

- Propose a novel approach for learning ensembles of neural networks.
- Combination of SG-MCMC sampling, group sparse prior and network pruning.
- Experimental verifications for sparse structure learning for LSTM models.

Future work:

- Interleaving model sampling and model pruning.
- Expand to more tasks.

THANK YOU FOR YOUR ATTENTION

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